Software Engineering Project Management

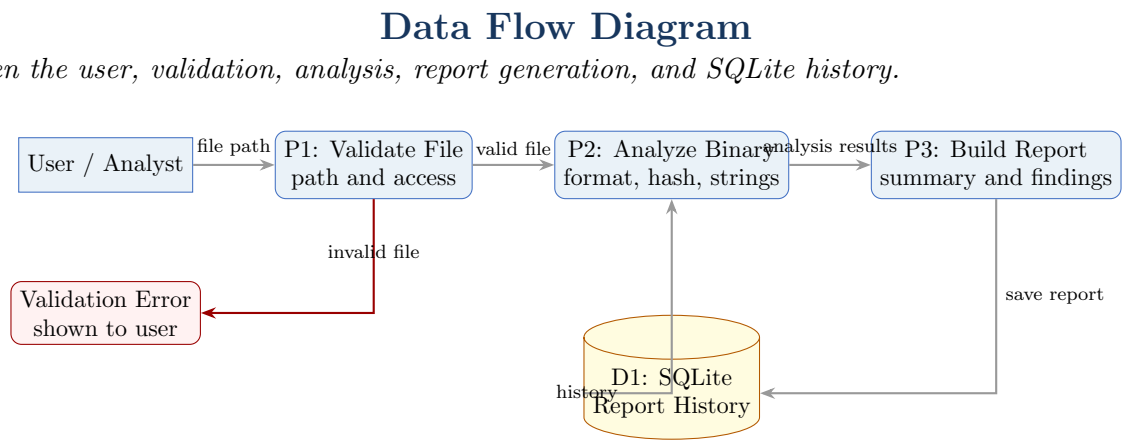
UML & System Design Diagrams

CyberLab: Static Binary Analysis Toolkit

Prepared by: MD. Sabbir Mohatab
Student ID: IT-24038
Department: Information and Communication Technology (ICT)

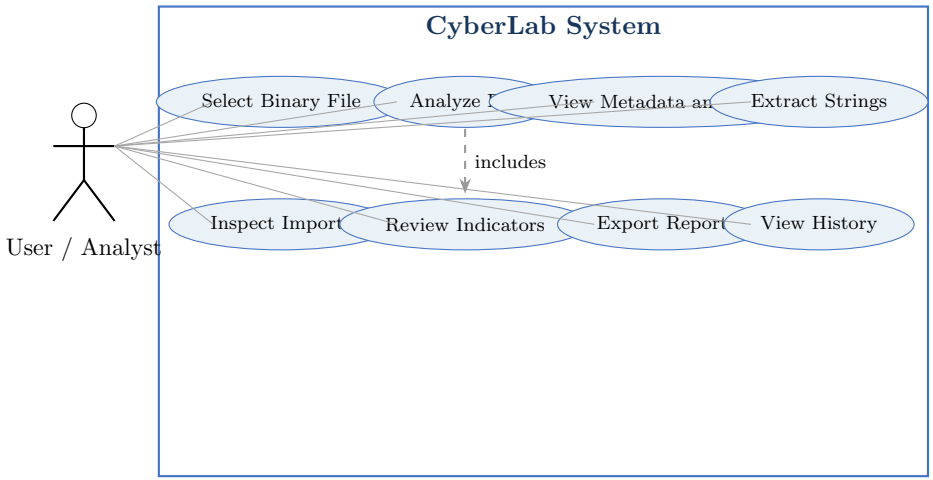
Includes DFD, Use Case, Activity, ERD, Class, and Object Diagrams

1. Data Flow Diagram (DFD — Level 1)



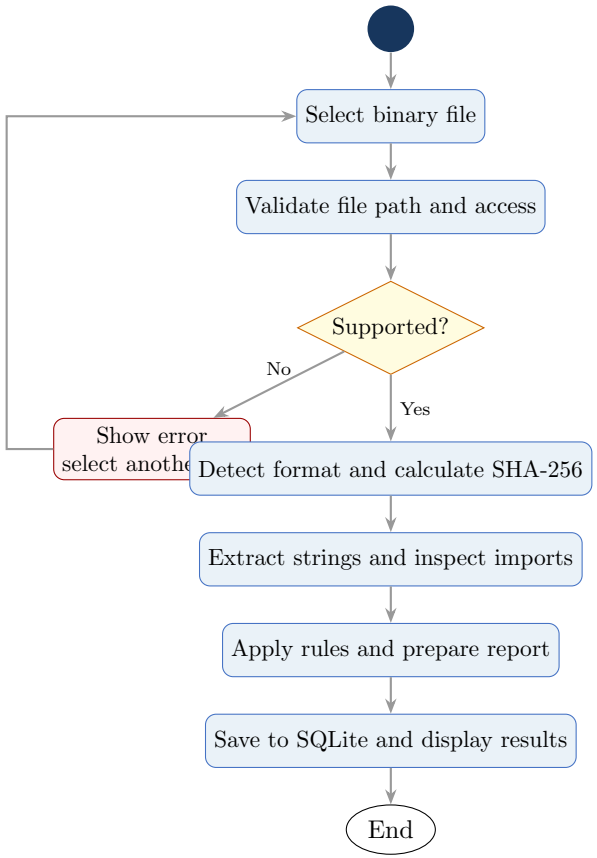
Shows the movement of data between the user, validation, analysis, report generation, and SQLite history.

2. Use Case Diagram



The user interacts with the CyberLab system to analyze files, inspect results, export reports, and view history.

3. Activity Diagram

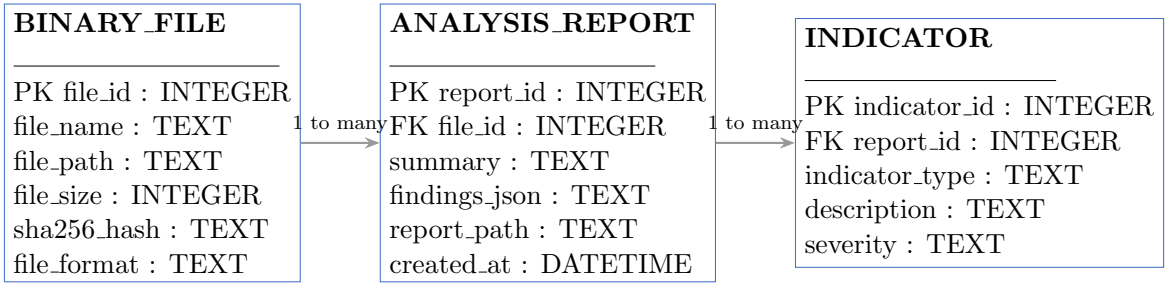


Workflow from file selection and validation to static analysis, report storage, and results.

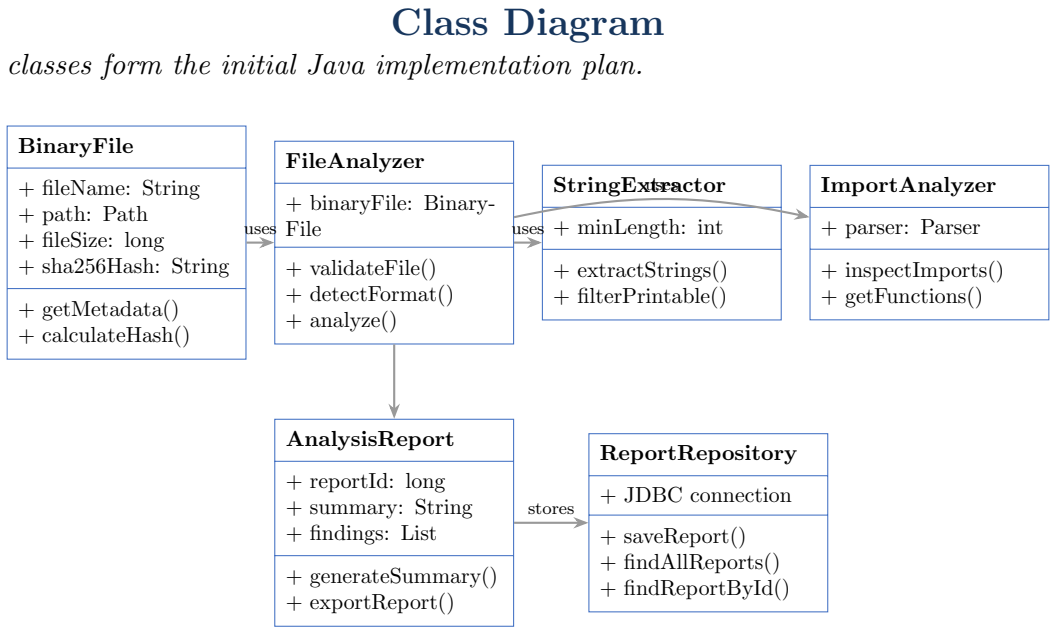
4. Entity Relationship Diagram (ERD)

ER Diagram

Proposed SQLite schema: a binary can have multiple reports, and each report can contain multiple indicators.

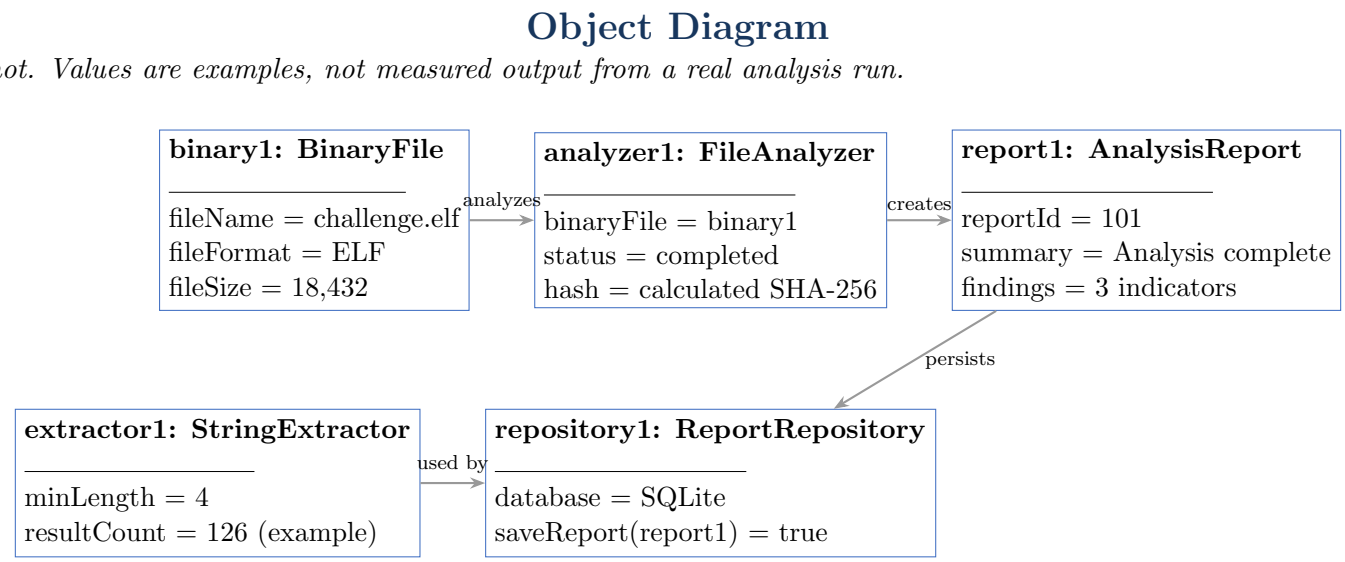


5. UML Class Diagram



Core object-oriented design. These six classes form the initial Java implementation plan.

6. UML Object Diagram



Illustrative runtime snapshot. Values are examples, not measured output from a real analysis run.